IMPLEMENTATION OF nice() SYSTEM CALL

In PureOS, the nice() system call can be implemented in two main ways: using a C program or directly through the terminal.

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1. Using nice() in a C Program

Here's a basic C program that uses nice() to change the process priority:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>

int main() {
    int current_nice = nice(0); // Get current nice value
    printf("Current nice value: %d\n", current_nice);

    nice(5); // Increase nice value by 5 (lower priority)

    int new_nice = nice(0); // Get new nice value
    printf("New nice value: %d\n", new_nice);

    // Simulate some work
    for (int i = 0; i < 100000000; i++);

    printf("Finished task with nice value: %d\n", new_nice);
    return 0;
}
```

Compile and run:

```
gcc nice_example.c -o nice_example
./nice_example
```

This program increases the nice value by 5, which means it lowers the priority of the process. The output shows how the nice value changes.

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2. Using nice and renice in the Terminal

You can also use the nice command directly from the terminal:

```
nice -n 10 ./my_program
```

This runs my_program with a nice value of 10 (lower priority).

To change the priority of a running process, use renice:

```
ps aux | grep my_program   # Find the PID
renice 5 -p <PID>          # Change the nice value to 5
```

Note: Only the root user can reduce the nice value (to raise priority).